

01_antonelli_vs_russell

June 13, 2026

1 Antonelli vs Russell — 2026 Qualifying, First Six Rounds (Chapter 1)

This is **Chapter 1** of a three-part look at one question: *how good is Kimi Antonelli, really, once you control for the car?* Here the control is the simplest one available — his **teammate**. Antonelli and George Russell drive the same Mercedes, so comparing their fastest qualifying laps isolates the driver. Chapters 2 (how he converts grids into wins) and 3 (cross-year track history) add two more independent ways to strip the car out.

This notebook compares both Mercedes drivers' fastest valid qualifying laps across the first six rounds of 2026, broken down by track segment. Round 5 (Canada) is where qualifying and the race result first diverged — Antonelli qualified *slower* than Russell yet won the race; Round 6 (Monaco) is the opposite extreme — pole by a clear margin and a flag-to-flag win. Caveats are listed at the end.

1.1 Method (briefly)

For each race: load qualifying, take each driver's fastest valid lap, and segment the track from FastF1's `circuit_info.corners` (corners within 250 m are grouped, straights are filled in between). Telemetry — including FastF1's per-sample `Time` channel — is resampled onto a uniform 5 m distance grid. Per-segment time is read directly from that resampled `Time` channel as `Time[last_step] - Time[first_step]` inside the segment. Per-segment delta is `time(Russell) - time(Antonelli)`, so **positive = Antonelli faster**.

An earlier version integrated `step / speed` per grid point. On 2 of 4 races the accumulated residual exceeded the 0.1 s sanity-check threshold (see the next section), so the analysis switched to reading FastF1's sample times directly. Residuals are now 0.1 s on all six races.

Each segment is classified into one of four categories by its minimum speed: slow corner (<130 kph), medium corner (130–200 kph), fast corner (>200 kph), or straight.

core	INFO	Loading data for Australian Grand Prix - Qualifying
[v3.8.1]		
req	INFO	Using cached data for session_info
req	INFO	Using cached data for driver_info
req	INFO	Using cached data for session_status_data
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req          INFO      Using cached data for _extended_timing_data
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req          INFO      Using cached data for race_control_messages
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req          INFO      Using cached data for session_info

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core         INFO      Loading data for Monaco Grand Prix - Qualifying [v3.8.1]
req          INFO      Using cached data for session_info
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```

	Round	Race	ANT lap (s)	RUS lap (s)	Δ lap (s)	ANT Q \
0	1	Australian Grand Prix	78.811	78.518	-0.293	3
1	2	Chinese Grand Prix	92.064	92.286	0.222	3
2	3	Japanese Grand Prix	88.778	89.076	0.298	3
3	4	Miami Grand Prix	87.798	88.197	0.399	3
4	5	Canadian Grand Prix	72.646	72.578	-0.068	3
5	6	Monaco Grand Prix	72.051	72.445	0.394	3

	RUS Q	Q mismatch
0	3	False
1	3	False

2	3	False
3	3	False
4	3	False
5	3	False

No Q-session mismatches across the 4 races.

1.2 Sensor quality

FastF1 telemetry doesn't carry an explicit quality flag, but a frozen `Speed` sensor produces a clear signature: only a handful of unique speed values across many samples within one segment. Any segment that fails this check is excluded from the headline category analysis below — the per-segment time delta is also unreliable in those segments because cumulative `Distance` is integrated from the bad speed.

Flagged 5 segment(s) where a driver's speed sensor looks frozen:

	race	segment	kind	category	min_speed_kph	\
0	Japanese Grand Prix	S7	straight	straight	177.2	
1	Japanese Grand Prix	C7	corner	medium_corner	189.0	
2	Japanese Grand Prix	S8	straight	straight	189.0	
3	Japanese Grand Prix	C8	corner	medium_corner	189.0	
4	Japanese Grand Prix	S9	straight	straight	97.0	

	delta_s_pre_filter
0	-3.555
1	-0.766
2	-1.663
3	-2.416
4	8.712

1.3 Sanity check

The segmentation logic is only useful if the segment deltas sum back to the actual lap-time delta. The table below verifies this on real telemetry: any $|\Delta|$ above 0.1 s would indicate a bug.

	Race	Σ segment Δ (s)	Lap-time Δ (s)	residual (s)
0	Australian Grand Prix	-0.3838	-0.293	0.09084
1	Chinese Grand Prix	0.2352	0.222	0.01318
2	Japanese Grand Prix	0.2758	0.298	0.02216
3	Miami Grand Prix	0.3758	0.399	0.02316
4	Canadian Grand Prix	-0.1352	-0.068	0.06718
5	Monaco Grand Prix	0.3938	0.394	0.00016

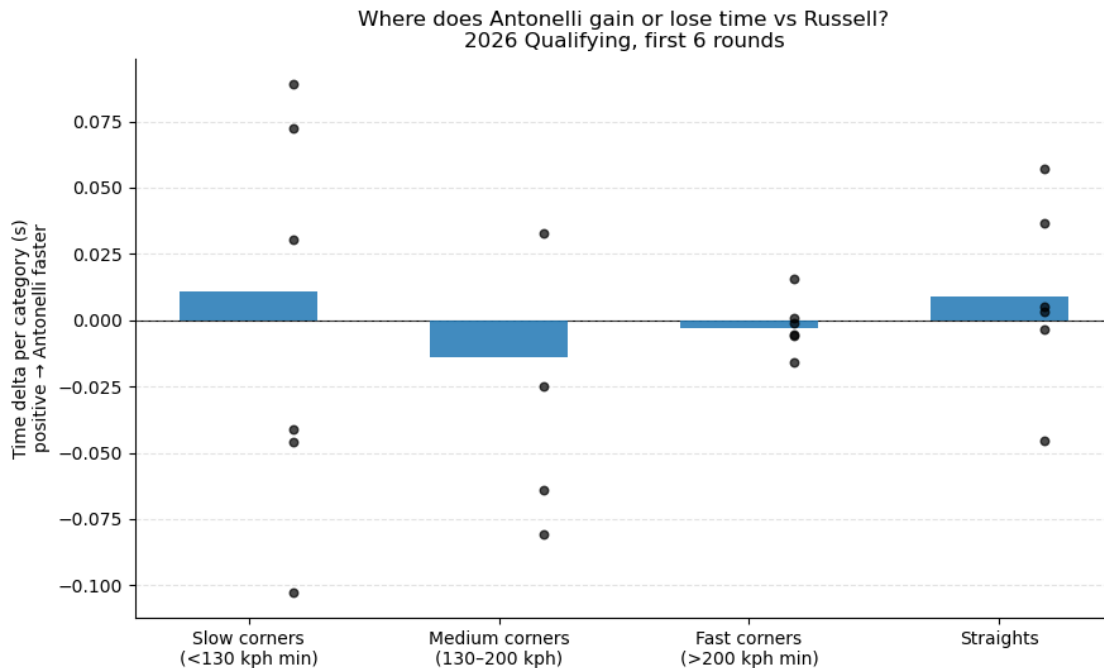
category	fast_corner	medium_corner	slow_corner	straight
event_name				
Australian Grand Prix	3	1	3	8
Canadian Grand Prix	1	2	3	7
Chinese Grand Prix	2	1	5	9

Japanese Grand Prix	2	5	1	9
Miami Grand Prix	2	0	4	7
Monaco Grand Prix	1	0	3	5

1.4 Headline finding — where does the time go?

Each bar is the mean per-category delta across all six races. Black dots show each individual race's per-category mean — they tell you whether the average reflects a consistent pattern or a noisy one.

Headline chart uses 79 segments after excluding 5 with sensor-freeze flags.



Per-category means after filtering (s/lap, positive = ANT faster):

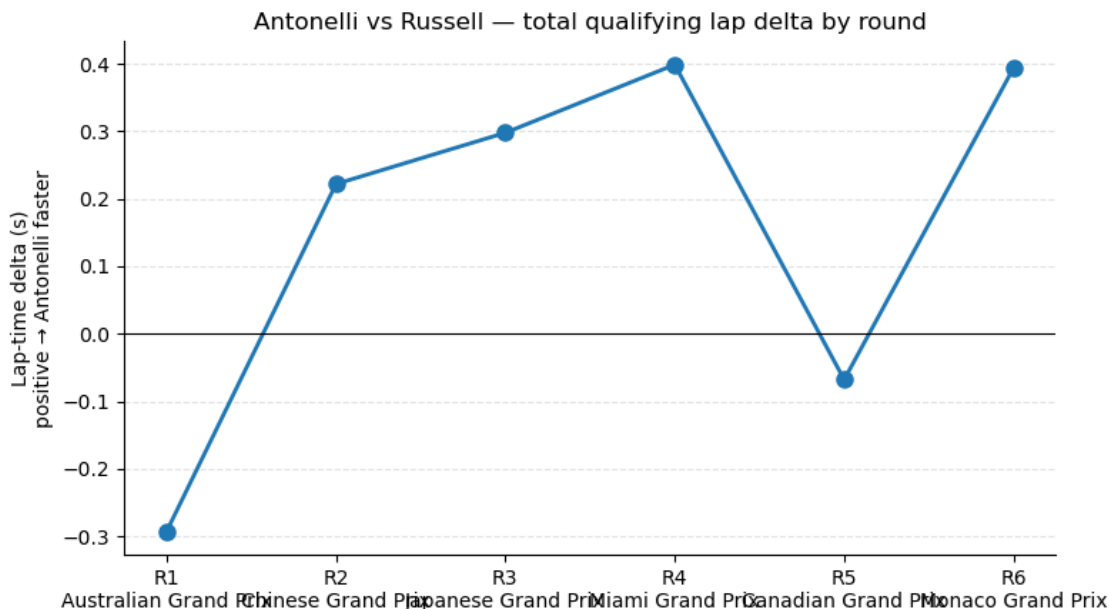
```
category
fast_corner      -0.0032
medium_corner    -0.0138
slow_corner       0.0110
straight         0.0089
```

The segment-level picture, after excluding five Japan segments where Antonelli's speed sensor was frozen for ~1300 m of the lap, is quieter than I expected — all four categories sit within ± 0.014 s/lap of zero:

- **Straights:** essentially tied ($+0.009$ s/lap).
- **Slow corners (<130 kph):** essentially tied ($+0.011$ s/lap).
- **Fast corners (>200 kph):** essentially tied (-0.003 s/lap).
- **Medium corners (130–200 kph):** essentially tied (-0.014 s/lap).

Whatever separates the two drivers, it isn't concentrated in one phase of the lap — it's spread thin across the whole thing. (The corner-cycle view further down finds where the real difference lives.)

1.5 Trajectory — is the gap closing?



Across six rounds the qualifying gap is **not** the clean monotone climb the first six rounds suggested. It went -0.29 s (Russell faster) at Australia $\rightarrow +0.22 \rightarrow +0.30 \rightarrow +0.40$ s (Antonelli's peak) at Miami, then **-0.07 s at Canada** (Russell briefly back ahead), then **$+0.39$ s at Monaco**. Laid out: R1 $-0.29 \rightarrow$ R2 $+0.22 \rightarrow$ R3 $+0.30 \rightarrow$ R4 $+0.40 \rightarrow$ R5 $-0.07 \rightarrow$ R6 $+0.39$, with a six-race mean of **$+0.16$ s** in Antonelli's favour. He out-qualifies Russell more often than not, and by a clear margin when he does — but Canada is a reminder that the trend is noisy at this sample size, and (as Chapter 2 shows) qualifying isn't the whole story of how he wins.

1.6 Where on the track is each driver gaining time?

The category breakdown averages across each segment, but it can hide where exactly within a segment the time is found. Below, each circuit is shown as a track map coloured by the **local slope of the cumulative time delta** — the rate at which the gap is changing as the lap unfolds.

- **Blue** = Antonelli is gaining on Russell at that part of the track.
- **Red** = Russell is gaining on Antonelli.
- Corner numbers (from FastF1's `circuit_info`) are overlaid for orientation.

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req           INFO      Using cached data for session_info

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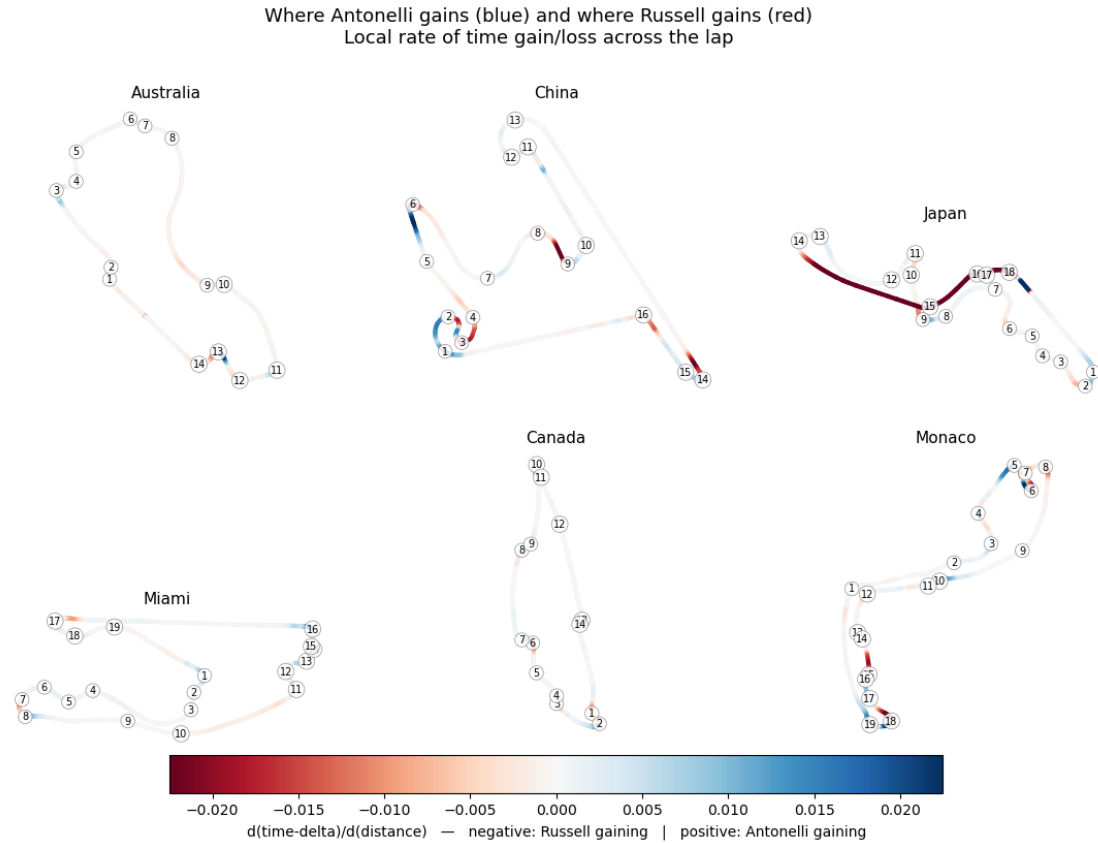
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```



1.7 How is he winning?

We know Antonelli is faster — at every track, every round. But how, specifically? The segment-level breakdown (above) turned out flat: straights, fast corners, slow corners, medium corners all essentially tied. So the advantage has to be hiding in something that segment-time averaging blurs over.

To find it, I looked at two specific signals at each corner on the lap. **Apex speed**: how much speed each driver carries through the deepest point of the corner. **Brake and throttle timing**: how late each driver brakes going in, and how soon they get back on throttle coming out.

1.7.1 Apex speed isn't where he's winning

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```

Apex-speed delta by corner-speed bucket (positive = Antonelli faster at the apex):

	mean ANT-RUS (kph)	n
speed_bucket		
very slow (<100)	-0.98	16
slow (100-150)	-0.43	22
medium (150-200)	-0.00	22
fast (>=200)	2.27	36

At slow and medium corners — the ones with a real apex where commitment matters — Antonelli is tied or slightly behind Russell. He's not carrying noticeably more speed through the deepest part

of the corner. The narrow positive at fast corners is real but small, and many of those “corners” in FastF1’s circuit info are really high-speed kinks where the apex notion doesn’t apply cleanly.

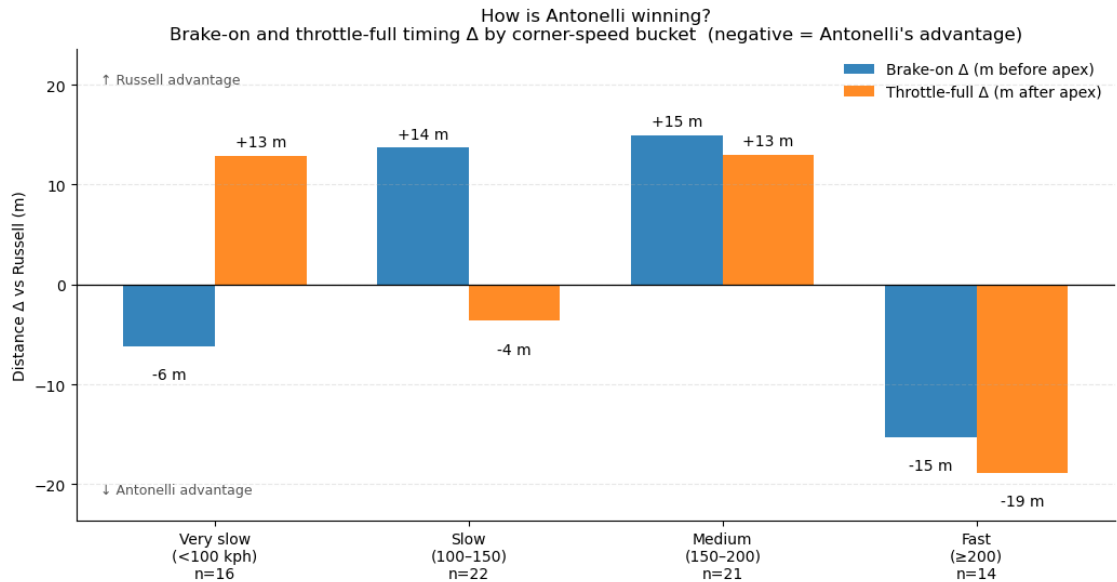
So raw apex commitment isn’t the answer. The advantage has to come from somewhere else in the corner cycle.

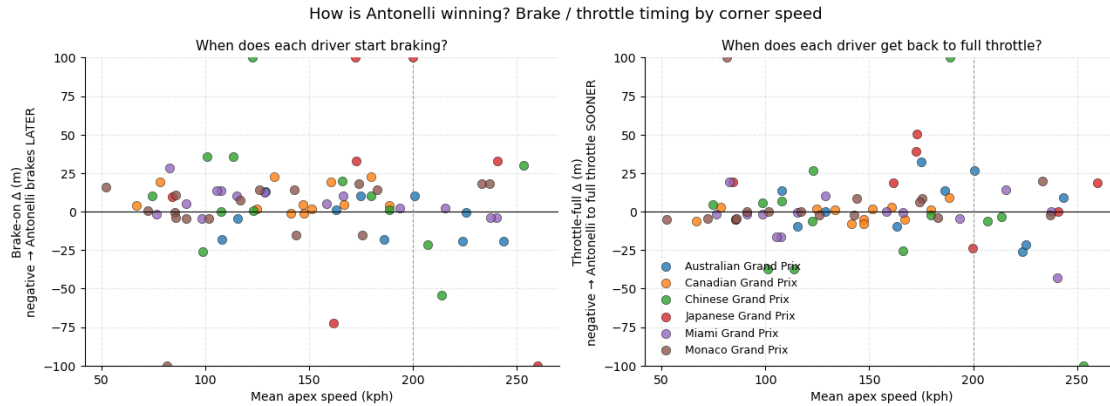
1.7.2 He brakes later and gets back on throttle sooner — but only at fast corners

Brake-on Δ (negative = Antonelli brakes LATER)

Throttle-full Δ (negative = Antonelli back on full throttle SOONER)

speed_bucket	brake_on_delta		throttle_full_delta	
	mean	count	mean	count
very slow (<100)	-6.2	16	12.9	16
slow (100-150)	13.7	22	-3.6	22
medium (150-200)	14.9	21	13.0	21
fast (>=200)	-15.3	14	-18.8	14





There it is. At fast corners — and only at fast corners — Antonelli brakes about 15 m later than Russell on average AND gets back to full throttle about 19 m sooner. That’s the late-brake / early-throttle commitment signature you’d expect from a young driver willing to push the entry and exit of high-speed turns.

At slower corners his approach is the opposite: he brakes *earlier* and gets back on the gas *later* than Russell. It’s almost like two different driving philosophies depending on speed range.

So the mechanism is: he wins at fast corners through commitment in the braking and traction zones, and gives a bit of time back at slow corners. The fast-corner gain wins out on net.

A real caveat: 44 corners across 4 races, only $n=12$ in the fast bucket, and a handful of dramatic outliers (Japan T8 alone has Russell braking 200 m earlier than Antonelli at a 260-kph kink). The fast-corner signature is consistent and large enough to feel real, but I’d want more clean races before calling it final.

1.8 Year-over-year: rookie vs sophomore

The within-2026 segment-level breakdown collapsed under the data-quality filter, but the lap-level comparison against 2025 — Antonelli’s actual rookie season at Mercedes — is far more striking. The cell below pulls the same six tracks from 2025 and plots both years on one chart.

At each of these tracks Antonelli has gained on Russell year-over-year, between **0.29 and 0.77 s** (mean **+0.55 s/track**). This is the most durable signal in the notebook: a clear, consistent compression of the gap to his teammate from one season to the next.

```
core      INFO      Loading data for Australian Grand Prix - Qualifying
[v3.8.1]

req       INFO      Using cached data for session_info

req       INFO      Using cached data for driver_info

req       INFO      Using cached data for session_status_data

req       INFO      Using cached data for track_status_data

req       INFO      Using cached data for _extended_timing_data
```



```

req          INFO      Using cached data for timing_app_data
core         INFO      Processing timing data...
core         WARNING    Fixed incorrect tyre stint information for driver '4'
req          INFO      Using cached data for car_data
req          INFO      Using cached data for position_data
req          INFO      Using cached data for weather_data
req          INFO      Using cached data for race_control_messages
core         INFO      Finished loading data for 20 drivers: ['4', '81', '1',
'63', '22', '23', '16', '44', '10', '55', '6', '14', '18', '7', '5', '12', '27',
'30', '31', '87']
core         INFO      Loading data for Chinese Grand Prix - Qualifying
[v3.8.1]
req          INFO      Using cached data for session_info
req          INFO      Using cached data for driver_info
req          INFO      Using cached data for session_status_data
req          INFO      Using cached data for track_status_data
req          INFO      Using cached data for _extended_timing_data
req          INFO      Using cached data for timing_app_data
core         INFO      Processing timing data...
req          INFO      Using cached data for car_data
req          INFO      Using cached data for position_data
req          INFO      Using cached data for weather_data
req          INFO      Using cached data for race_control_messages
core         INFO      Finished loading data for 20 drivers: ['81', '63', '4',
'1', '44', '16', '6', '12', '22', '23', '31', '27', '14', '18', '55', '10',
'87', '7', '5', '30']
core         INFO      Loading data for Japanese Grand Prix - Qualifying
[v3.8.1]
req          INFO      Using cached data for session_info
req          INFO      Using cached data for driver_info
req          INFO      Using cached data for session_status_data
req          INFO      Using cached data for track_status_data
req          INFO      Using cached data for _extended_timing_data
req          INFO      Using cached data for timing_app_data

```

```

core          INFO      Processing timing data...
req           INFO      Using cached data for car_data
req           INFO      Using cached data for position_data
req           INFO      Using cached data for weather_data
req           INFO      Using cached data for race_control_messages
core          INFO      Finished loading data for 20 drivers: ['1', '4', '81',
'16', '63', '12', '6', '44', '23', '87', '10', '55', '14', '30', '22', '27',
'5', '31', '7', '18']
core          INFO      Loading data for Miami Grand Prix - Qualifying [v3.8.1]
req           INFO      Using cached data for session_info
req           INFO      Using cached data for driver_info
req           INFO      Using cached data for session_status_data
req           INFO      Using cached data for track_status_data
req           INFO      Using cached data for _extended_timing_data
req           INFO      Using cached data for timing_app_data
core          INFO      Processing timing data...
req           INFO      Using cached data for car_data
req           INFO      Using cached data for position_data
req           INFO      Using cached data for weather_data
req           INFO      Using cached data for race_control_messages
core          INFO      Finished loading data for 20 drivers: ['1', '4', '12',
'81', '63', '55', '23', '16', '31', '22', '6', '44', '5', '7', '30', '27', '14',
'10', '18', '87']
core          INFO      Loading data for Canadian Grand Prix - Qualifying
[v3.8.1]
req           INFO      Using cached data for session_info
req           INFO      Using cached data for driver_info
req           INFO      Using cached data for session_status_data
req           INFO      Using cached data for track_status_data
req           INFO      Using cached data for _extended_timing_data
req           INFO      Using cached data for timing_app_data
core          INFO      Processing timing data...
req           INFO      Using cached data for car_data
req           INFO      Using cached data for position_data

```

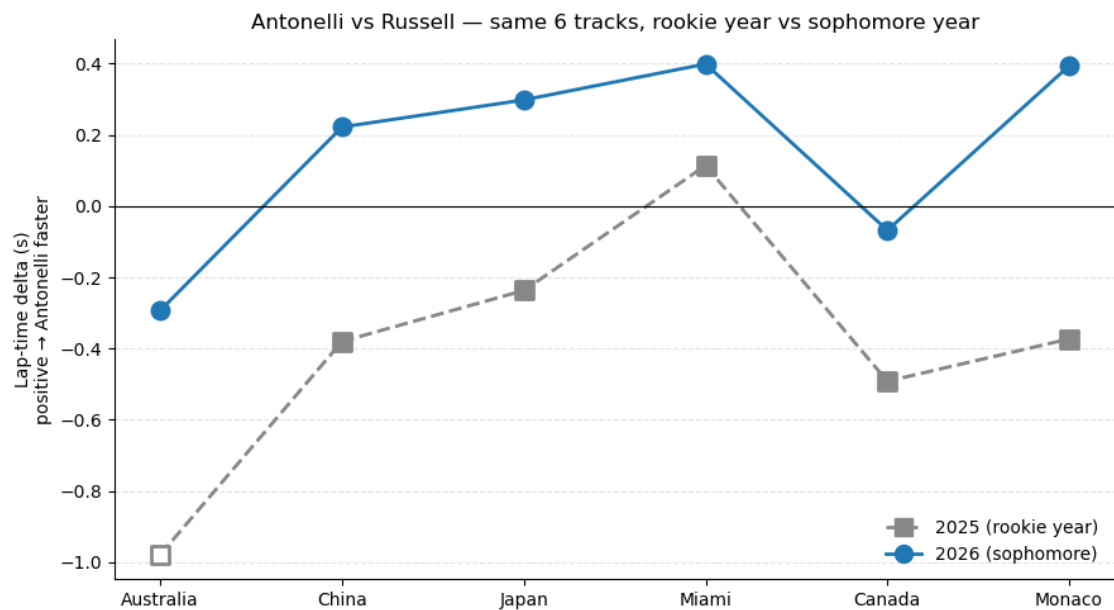
```

req          INFO      Using cached data for weather_data
req          INFO      Using cached data for race_control_messages
core         INFO      Finished loading data for 20 drivers: ['63', '1', '81',
'12', '44', '14', '4', '16', '6', '23', '22', '43', '27', '87', '31', '5', '55',
'18', '30', '10']
core         INFO      Loading data for Monaco Grand Prix - Qualifying [v3.8.1]
req          INFO      Using cached data for session_info
req          INFO      Using cached data for driver_info
req          INFO      Using cached data for session_status_data
req          INFO      Using cached data for track_status_data
req          INFO      Using cached data for _extended_timing_data
req          INFO      Using cached data for timing_app_data
core         INFO      Processing timing data...
req          INFO      Using cached data for car_data
req          INFO      Using cached data for position_data
req          INFO      Using cached data for weather_data
req          INFO      Using cached data for race_control_messages
core         INFO      Finished loading data for 20 drivers: ['4', '16', '81',
'44', '1', '6', '14', '31', '30', '23', '55', '22', '27', '63', '12', '5', '87',
'10', '18', '43']

```

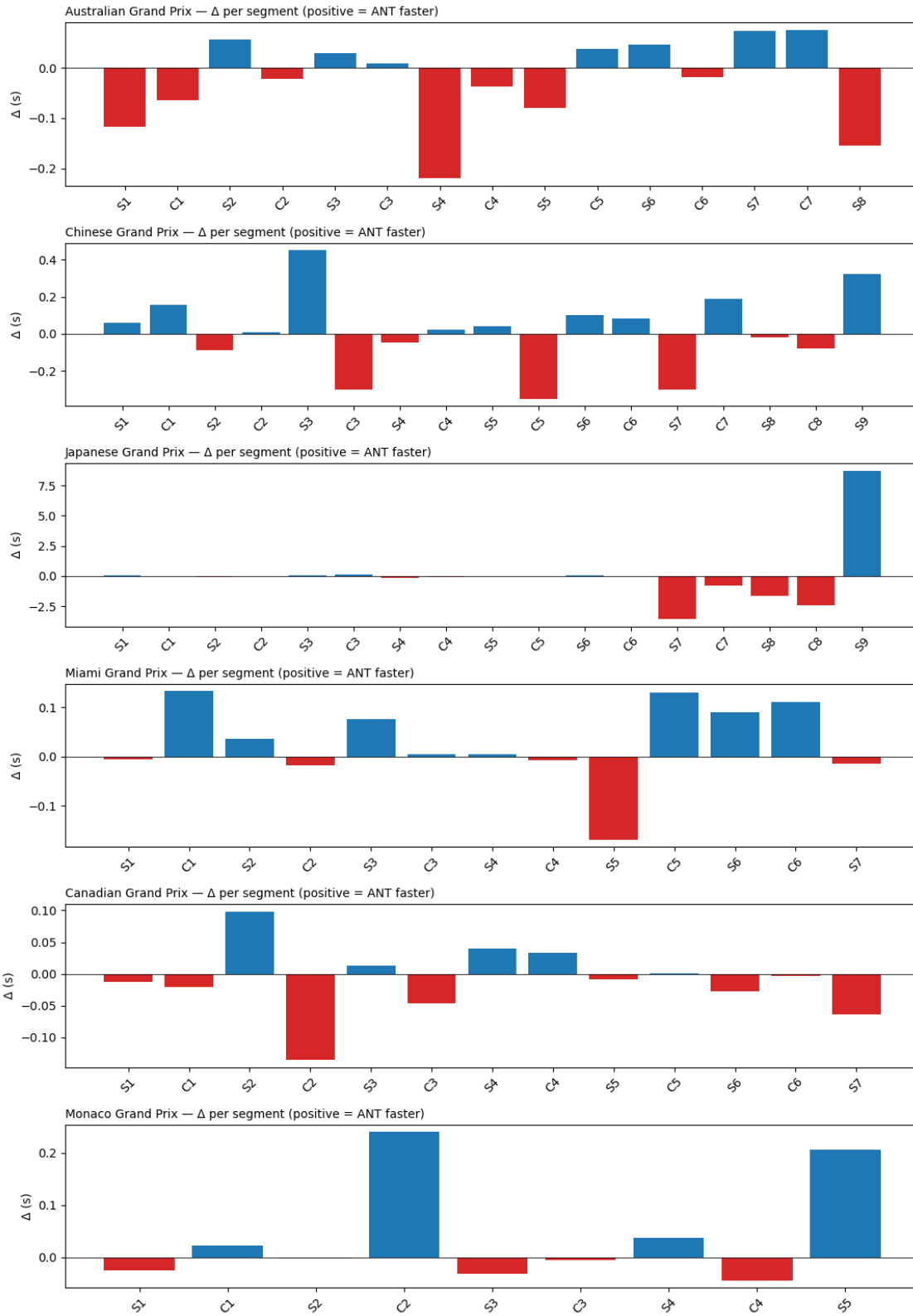
	Track	2025 Δ (s)	2026 Δ (s)	YoY gain (s)	2025 Q-mismatch?
0	Australia	-0.979	-0.293	0.686	True
1	China	-0.380	0.222	0.602	False
2	Japan	-0.237	0.298	0.535	False
3	Miami	0.114	0.399	0.285	False
4	Canada	-0.492	-0.068	0.424	False
5	Monaco	-0.373	0.394	0.767	False

Mean YoY gain across the 4 tracks: +0.550 s



1.9 Per-race detail (appendix)

For each race, the raw per-segment delta. Useful for curious readers; not the main story.



1.10 Caveats

- **Sample size: six races.** Findings are directional, not conclusive — and as the sensor investigation above shows, segment-level findings in particular are fragile to single-race data issues.
- **Q-session timing.** Q1 / Q2 / Q3 happen on an evolving track. The summary table flags any race where the two drivers’ fastest valid laps came from different Q-segments.
- **Setup divergence.** Mercedes drivers do not always run identical setups. Public telemetry can’t separate driver from setup.
- **Traffic and tire age within Q.** Out-laps, in-laps, and intra-session tire age all affect achievable lap time.
- **Telemetry sensor freezes.** FastF1’s car telemetry occasionally stops reporting changes for long stretches. In this dataset, five Japan segments (S7, C7, S8, C8, S9 — covering D 4000 m to the end of the lap) are excluded by the sensor-quality filter above. Lap-level and sector-level deltas are unaffected — those come from timing-line beams, independent of the car telemetry. Expect mild visual distortion at Japan’s T14–T18 in the track-delta map.

1.11 What I found

The strongest signal across six races is the **year-over-year comparison**. Antonelli has gained between 0.29 and 0.77 s on Russell at each of the same six tracks since his rookie season — mean **+0.55 s/track**. He started his rookie year well behind Russell and has compressed that gap markedly in his sophomore season.

Within 2026, he out-qualifies Russell more often than not (six-race mean **+0.16 s**), but the round-by-round trajectory is noisy rather than monotone: he was behind at Australia, peaked at Miami, slipped behind again at Canada, then took a dominant Monaco pole. The segment-level split is essentially flat; the one place a concrete mechanism shows up is the fast-corner brake/throttle timing.

That’s qualifying — getting to the front. **Chapter 2** asks how those grids become wins, and **Chapter 3** asks whether his strong tracks are driver tracks or car tracks.